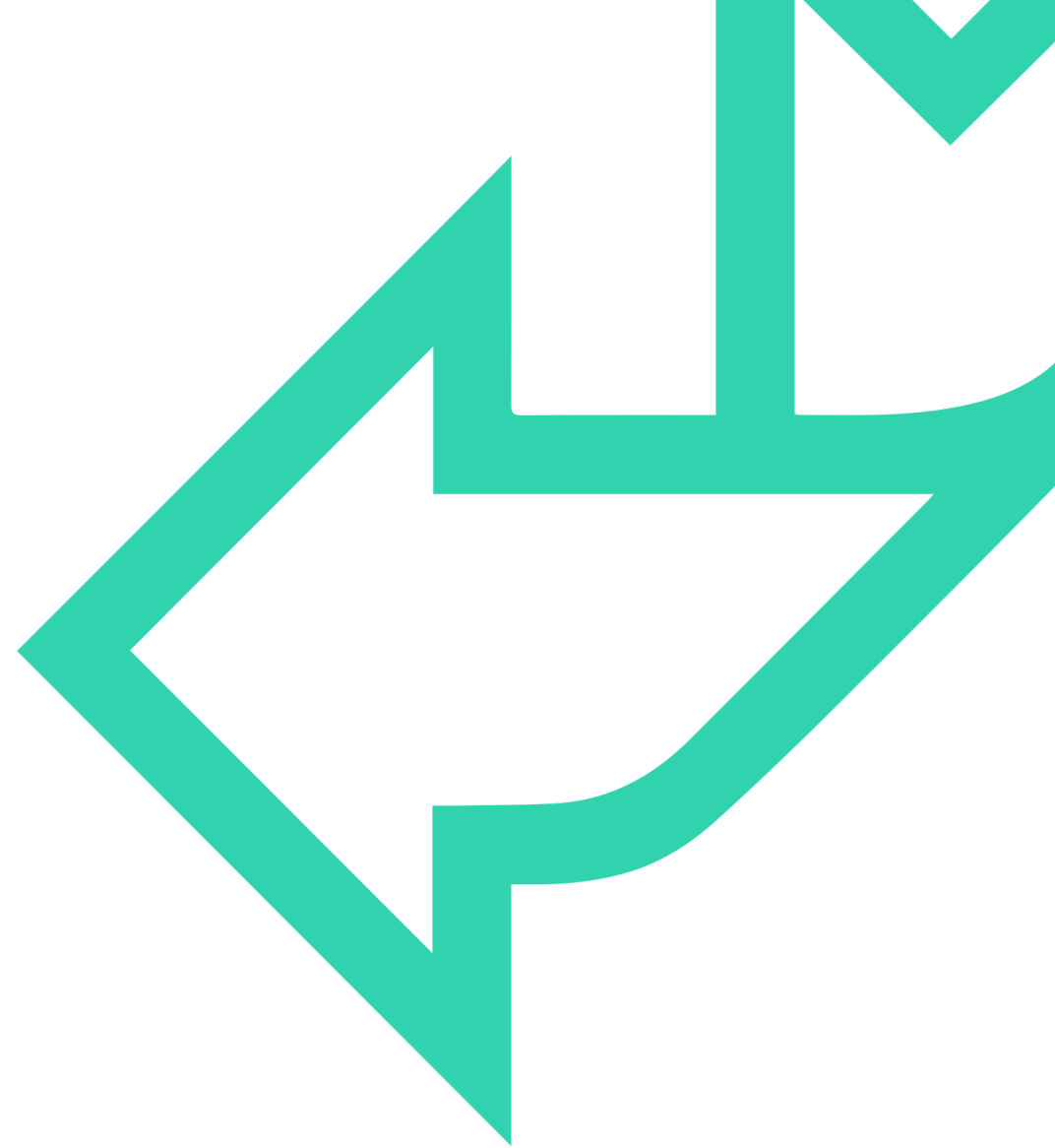




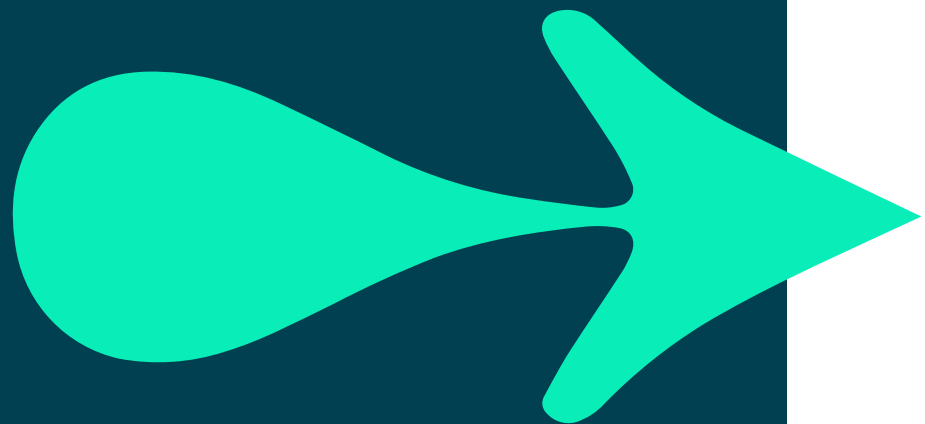
Docker Compose





WHAT IS DOCKER COMPOSE

- **Can create collections of containers**
 - On the same server
 - Creates a network between them so they can call each other using their service name
- Enables you to define and run multiple containers with a single command.
- Uses a single configuration file to specify deployment
- Is a great tool for spinning microservice apps



Why use Docker Compose?

- **Ease of Use:** `docker compose up/down`
- **Simplified Multi-Container Management:** `compose.yml`

`docker-compose.yml` (or `.yaml`)
Is also acceptable but best use `compose.yml`
- **Environment Consistency**
 - Consistency between Dev, test, and production environments by using the same configuration.
- **Simplified Networking**
 - creates a network for your services, allowing them to communicate with each other using simple service names.
- **Service Dependency Management** ensuring that services start in the correct order.
 - Useful for databases, message queues, or any other service that must be up and running before other services start.
- **Persistent Data:**
 - Can define and manage data volumes, ensuring data persistence and data sharing between containers.

Compose CLI

docker compose up

Start up all the services declared in the configuration

- d** Run the containers in the background
- build** Build images before starting containers
- scale [SERVICE]=[NUM]** Can specify the number of containers

docker compose down

Stops the services created up.

- rmi** Removes images.

docker compose build

Builds the images without running any containers.

- parallel** Builds all the images at the same time to save time when images do not depend on each other

Note: **docker-compose** set of commands are deprecated

Compose CLI

docker compose ps

Displays the running containers started by Docker Compose

-a Show all containers, including stopped ones

docker compose exec

Execute a command in a running container

-d Run the command in the background

-e Set environment variables

docker compose logs

Display the log outputs from the running services

docker compose push

Push the images used for the services to a registry



LAB

Please do the following lab

4-Docker Compose.docx

